

Voice Order Support Agent

Interrupt it any time. It **stops at once** — and remembers **only what you heard**.

Live: voice-order-support-agent--my5757980.replit.app

Code: github.com/my5757980/voice-order-support-agent · MIT

“Where is my order?” is the question support answers most — and needs a human for least.

- Post-purchase support is the **highest-volume, lowest-value** contact in e-commerce. The answer is already in data the business owns.
- Today a shopper gets a **queue**, an **IVR menu**, or a **chatbot** that collapses on the first follow-up.
- The real need is conversational — pronouns, corrections, “no wait, the *other* order”. That is a voice problem.
- Most voice agents make it worse: **you cannot interrupt them.**

WHAT IT DOES

Talk to it like a person at the counter.

Looks it up first

Asks nothing it can find out. Lists your orders before asking which one.



Answers briefly

Two sentences. Tracking numbers and order references go **on screen**, never read aloud.



Acts carefully

Returns, cancellations, address changes: checks what is possible, reads it back, waits for an **explicit yes**.

Discloses it is automated in its first sentence. Hands over to a human on request.

We took Realtime STT, not the managed Voice Agent API — so we own turn-taking.

AssemblyAI Universal-Streaming

v3 WebSocket, 50 ms PCM16 frames, balanced turn detection — AssemblyAI's documented preset for customer support.

Streaming partials, every word

Barge-in and speculation read words as they are recognised, not only once final.

Our own orchestration

One asyncio task group per turn: cancelling an interrupted turn cancels everything below it, structurally.

Speculative dispatch

The model starts during the shopper's trailing silence — and the registry makes that turn read-only.

When the shopper talks over it, three things happen.

- **The browser's own playback buffer is zeroed** on the next audio render quantum — independently of the server stopping synthesis. Nothing else stops speech already buffered.
- **The first real word interrupts** — not the end of the sentence. "mhm", "okay", "yeah" never do.
- **Memory keeps only what was heard**: computed from the audio frames the browser actually played, mapped onto the reply — never what the model generated.

1.14 s

shopper starts talking → agent silent
recorded run, deployed app

BUILT TO BE TRUSTED WITH ACTIONS

Safety lives in the tool registry, not in the prompt.

Four gates before any call runs

Known tool · valid arguments · not a speculative turn · confirmation held. A model that skips the ritual still cannot act.

Consent, recognised properly

From the system's own refusal or the read-back; "yes, but..." and "wait" never count as yes.

Idempotent by construction

A database `UNIQUE` constraint, not an application check. Asking twice returns the same return.

Never claims a success

...a tool did not report. And never offers what the order does not allow.

MEASURED, AND SHOWN

Every turn draws its own latency waterfall.

First token · tool · first audio · end to end — on screen, per turn, from spans recorded server-side.

3.0 s

shopper stops → agent's voice
median, recorded run

1.14 s

talked over → silent
same run

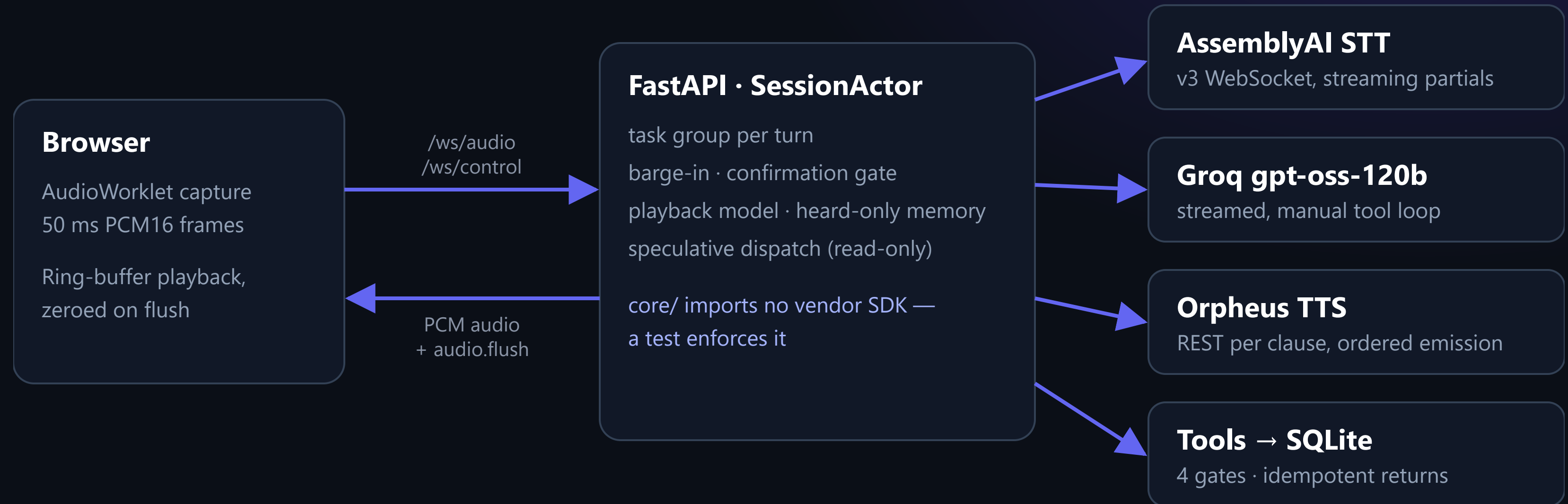
Honest caveat: Groq's free tier caps the model at 8,000 tokens and the voice at 10 requests a minute — the live demo serves one conversation at a time.

A clean test suite passed through all of these. Recording real conversations did not.

- Replies longer than four seconds **lost their first words** — a ring buffer overrun, measured: 8 s in, 4.06 s played.
- Barge-in was **impossible while a finished reply was still playing** — now the server models the browser's playback.
- Partial used finalized words only: the agent stopped **3.0 s** after being interrupted. Now about a second.
- "Do you want to cancel it?" **did not arm the confirmation**; a reasoning budget left turns **silent**. Both fixed, both tested.
- The voice **cut out on the second turn** with 90 of 100 daily requests unused: a 10-a-minute cap the headers never show. A reply is now two requests, not one per clause.

ARCHITECTURE

One process, two sockets, ports and adapters.



TRY IT

Talk to it. Then talk over it.

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github.com/my5757980/voice-order-support-agent · MIT

AssemblyAI Universal-Streaming · Groq gpt-oss-120b · Canopy Labs Orpheus · FastAPI · AudioWorklet · 279 tests